C150: an exactly solved quantum cellular automaton

…and exactly what the solution does not determine

Krylov sector analysis, Tier 1 — report R8

The rule, and one naming note

- **C150 = Wolfram 150 = local symbol table (I, V, V, I)**
  - Hadamard fires at site i $\iff$ $x(i-1) \oplus x(i+1) = 1$
  - — the firing condition is exactly classical ECA-90's update
- **Boundary: obc0, the missing neighbour reads 0**
- **Cycle: brick-wall — even sublattice first, then odd**
- **Same object as HSF rule 6 with gate H (checked from HSF/src/model.jl)**
  - — so the whole Julia eigendata set is a usable independent oracle
- **Why this rule**
  - R2 saw binomial sector sizes; R1 uses it as the HSF regression fixture;
  - and its fitted growth exponent 2.022 sat above the informational ceiling 2

Five results

- **1. Reduction — sector partition = a single-flip graph**
  - for ANY unitary rule; the brick-wall ordering drops out
- **2. Solution — walls, hard-core hopping, closed form**
  - $|S_w| = C(N+1, w)$ for even w; the charge is the Ising domain-wall number
- **3. C150 is symmetry-resolved, NOT fragmented**
  - sector count is linear in N — it is the family's null model
- **4. Interacting, but not chaotic**
  - #distinct eigenvalues = $3^{\lceil N/2 \rceil}$ exactly — no RMT ensemble fits
- **5. Area law vs volume law is a sector statement**
  - …but the sector fixes only the ceiling, not the value

Result 1 — the sector partition is a flip graph

- **Theorem. For a unitary rule the sectors are the connected components of**
  - $x \sim x \oplus 2^i$ for every site i where the Hadamard fires
- **Proof sketch**
  - each half-layer is depth-1 ⇒ its image from $|x\rangle$ is a subcube (R1 no-interference)
  - x lies in its own image ($\langle b|H|b\rangle \neq 0$) ⇒ composite image ⊇ both half-layer images
  - inside a half-layer the V/I pattern is frozen ⇒ the subcube is single-flip connected
- **Two consequences**
  - the brick-wall ORDER is irrelevant — the partition is a property of the symbol table
  - cost drops from $O(2^N \cdot |succ|)$ to $O(N \cdot 2^N)$ — this is what reaches N = 32
- **Validated: all 16 unitary rules × both boundaries × N = 3…12 — 320 units, 0 mismatches**

Result 2 — walls: the whole rule in one line

- **Bond variables $b_j = x_j \oplus x_{(j+1)}$, $j = 0\ldots N$, $x(-1) = x(N) = 0$**
  - $x \mapsto b$ is a BIJECTION onto the even-weight strings of length N+1
- **The elementary move**
  - flipping x_i toggles bonds b_i, $b_{(i+1)}$, and is allowed iff $b_i \neq b_{(i+1)}$
  - ⇒ a HARD-CORE HOP of one domain wall. Walls are never created or destroyed.
- **Therefore**
  - conserved charge $Q = \Sigma\, (1 - Z_j Z_{(j+1)})/2$ — the open Ising domain-wall number
  - each wall shell is exactly one sector (hard-core hopping is connected on a shell)
- **Zero charge violations over every state, N = 3…12, both boundary conditions**

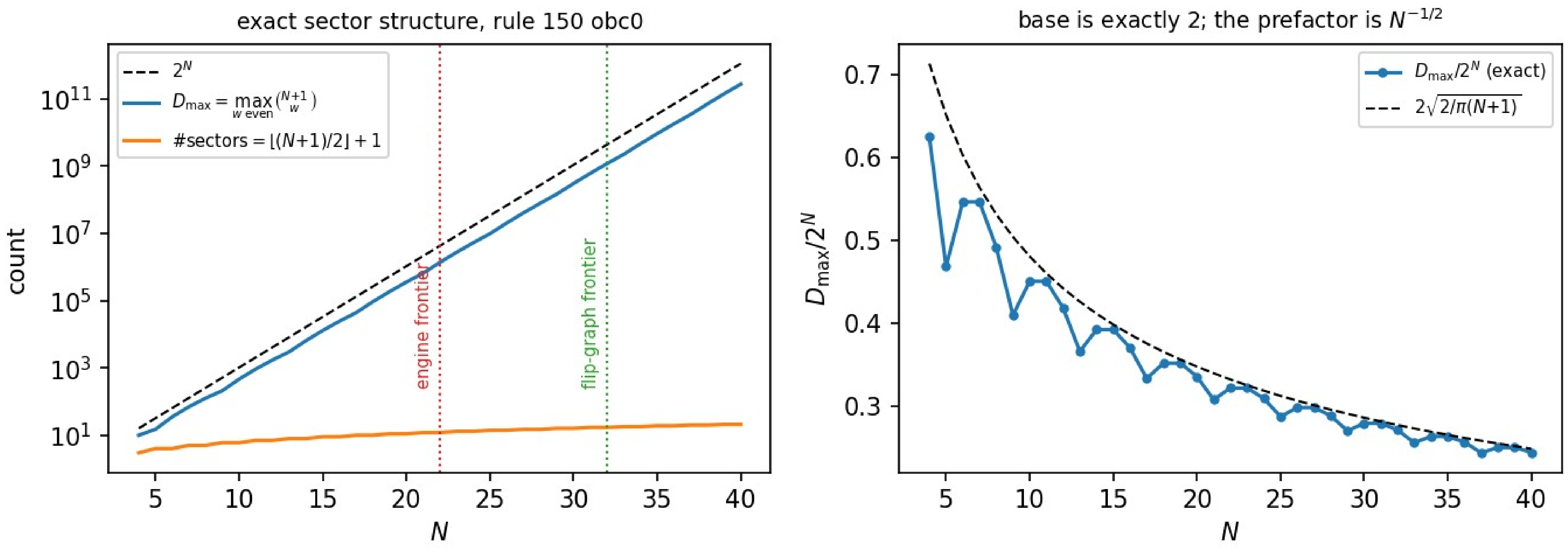
Result 3 — the closed form, and the growth law

- **$|S_w| = C(N+1, w)$ for even w   #sectors = $\lfloor (N+1)/2 \rfloor + 1$ D_{max} = max over even w**
  - matches every stored Tier-1a unit, N = 6…20, both boundary conditions
- **C150 is NOT fragmented**
  - sector count is LINEAR in N and every sector is a full Q-eigenspace
  - no Krylov structure beyond the U(1) charge — contrast W204's 2^N frozen singletons
  - → it is the wrong example to quote for Hilbert-space fragmentation
- **The 2.022 anomaly, resolved**
  - $D_{max}/2^N = 2\sqrt{2/\pi(N+1)}$ — base is EXACTLY 2, the decay is an $N^{-1/2}$ prefactor
  - plus a sawtooth at $N \equiv 1 \pmod 4$, where the central binomial sits at odd w
  - a pure-exponential fit reads prefactor + sawtooth as a base above the ceiling

Numerical frontier — two independent routes

N	2^N	#sectors	D_max	streamed engine	flip graph
20	1,048,576	11	352,716	1.1 h	—
21	2,097,152	12	646,646	3.6 h	0.3 s
22	4,194,304	12	1,352,078	10.2 h	0.4 s
28	268,435,456	15	77,558,760	—	37.7 s
30	1,073,741,824	16	300,540,195	—	154 s
31	2,147,483,648	17	601,080,390	—	297 s
32	4,294,967,296	17	1,166,803,110	—	815 s

Exact sector structure, and where the numerics stop

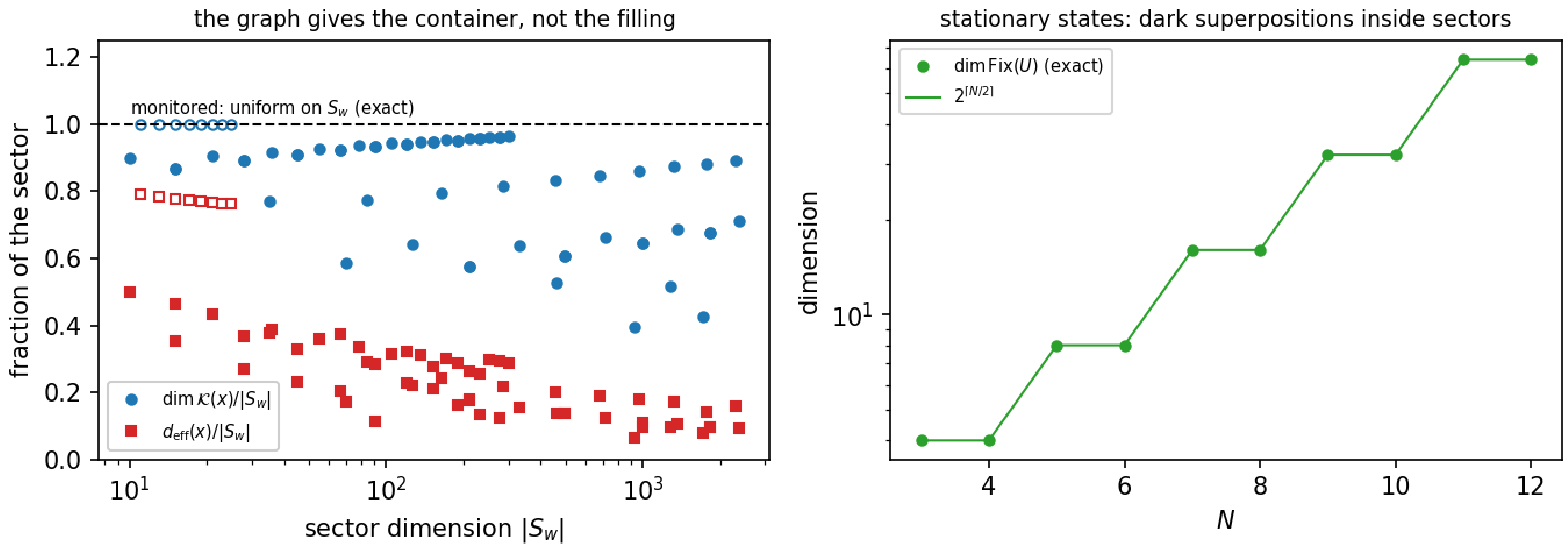


Left: the closed form with both computed frontiers marked. Right: D_max/2^N against 2√(2/π(N+1)) — base 2, N^(−1/2) prefactor, sawtooth at N ≡ 1 (mod 4).

What the sector does NOT determine

- **A sector is the maximal reachable set from a basis state — nothing more**
- **An exponentially large stationary space**
  - dim $Fix(U_w) = C(\lceil N/2 \rceil, w/2)$, dim $Fix(U) = 2^{\lceil N/2 \rceil} = 2^{(\#\text{even sites})}$
  - $Fix(U) = Fix(L_A) \cap Fix(L_B)$; almost all of it is DARK SUPERPOSITIONS
- **Hence an inner sector is never a Krylov space**
  - spectrum degenerate $\Longleftrightarrow |S_w| > N+1$; dim $K(x)/|S_w| = 0.40 \ldots 0.96$
- **Monitored vs unmonitored disagree inside a sector**
  - monitored: provably uniform on S_w (doubly stochastic, irreducible, aperiodic)
  - unmonitored: diagonal ensemble occupies only 6.6 % … 50 % of the sector
- **Reflection: commutes for odd N; conjugates U to U^{-1} for even N**

The graph gives the container, not the filling



Left: Krylov and diagonal-ensemble fractions of a sector; the dashed line is the monitored answer, which is exactly uniform. Right: dim Fix(U) = 2^[N/2].

Result 4a — interacting, not free

- **In the bond basis the gate at site i touches bonds i, i+1 only**
  - identity when both bonds agree; on the one-wall subspace a Hadamard coin
  - $M(c) = (1/\sqrt{2})[[-(-1)^c, 1], [1, (-1)^c]]$, $\det M = -1$, c = wall parity to the left
  - this bond circuit reproduces the engine to 1.1×10^{-16} — it IS C150
- **Two diagonal signs spoil gaussianity**
  - $\det(H) = -1$ but the gate acts by $+1$ on a doubly occupied pair → maximal nearest-neighbour wall interaction $\exp(i\pi\, n\, n)$
  - the $(-1)^c$ string sits on the DENSITY term, not the hop → not absorbed by Jordan–Wigner
- **Sharp test: a Gaussian circuit has additive many-body eigenphases**
  - C150: 0.55 … 1.87 rad   remove one defect: still 0.43 … 1.10 rad
  - remove BOTH: additive to 4×10^{-15} — so both signs are needed, and C150 is interacting

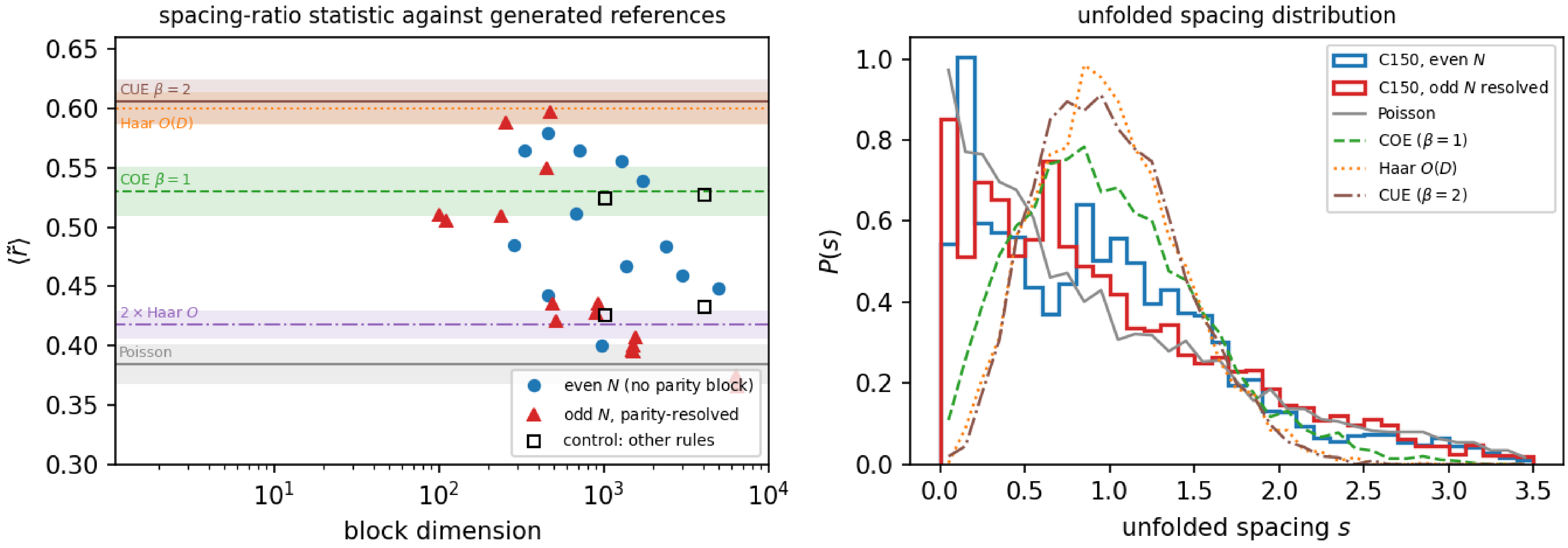
Result 4b — the spectrum is exponentially degenerate

N	2^N	#distinct	closed form 3^m / (3^m±1)/2	fraction
8	256	81	81	0.316
9	512	121	121	0.236
10	1,024	243	243	0.237
11	2,048	365	365	0.178
12	4,096	729	729	0.178
13	8,192	1,093	1,093	0.133
14	16,384	2,187	2,187	0.133

Result 4c — no random-matrix ensemble fits

- **β = 2 (GUE / CUE) excluded by symmetry**
  - U is real orthogonal ⇒ complex conjugation is a time reversal, $K^2=1$
  - ⇒ eigenvalues in mirror pairs; all statistics on $\theta \in (0, \pi)$ only
- **β = 1 (COE / Haar-O) excluded by the degeneracy law**
  - no chaotic Floquet operator has 3^m distinct eigenvalues out of 2^N
- **Poisson excluded too**
  - ONE universal set Θ_N per N: every shell's spectrum ⊆ Θ_N, and the largest shell realises ALL of it (small shells as little as 1.8 %)
  - largest shell N=16, w=8: $\langle \tilde{r} \rangle = 0.347$, BELOW Poisson (0.384) — and superposition can only approach Poisson from above
- **Structural source: U = L_B·L_A is a product of two involutions**
  - gives the ±θ pairing and explains Fix(U) = Fix(L_A) ∩ Fix(L_B)
- **⇒ C150 is INTERACTING BUT NOT CHAOTIC — the two are compatible**

Level statistics against generated references



Left: ⟨r̃⟩ per symmetry block with the reference ensembles as bands. Right: the unfolded spacing distribution.

Independent check: the HSF Julia eigendata (rule 6 = 150)

N	states	sectors	#distinct	3^m law	sizes	dim Fix	deg. states
8	256	5	81	81	✓	✓	0.43
9	512	6	121	121	✓	✓	0.53
10	1,024	6	243	243	✓	✓	0.56
11	2,048	7	365	365	✓	✓	0.67
12	4,096	7	729	729	✓	✓	0.66
13	8,192	8	1,093	1,093	✓	✓	0.73
14	16,384	8	2,187	2,187	✓	✓	0.74

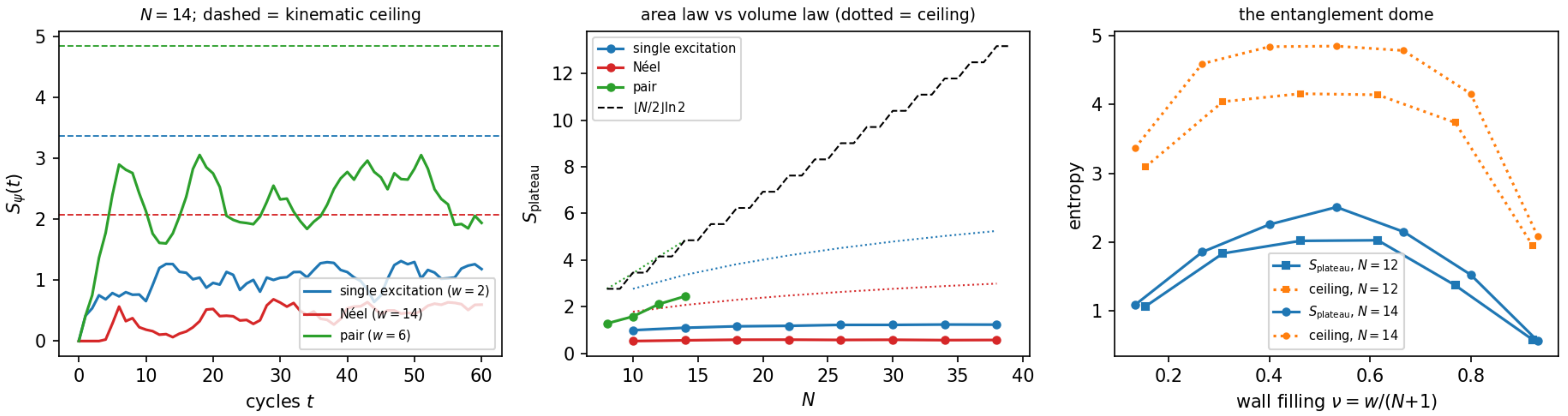
A caveat for the stored entanglement entropies

- **On a degenerate eigenvalue the eigensolver returns an ARBITRARY orthonormal basis**
  - and the half-chain entropy is not constant on that basis
- **Measured**
  - rotating inside one eigenspace moves the stored entropies by up to 0.27 nats
  - the same test on a simple eigenvalue gives a spread of exactly 0
- **For this rule that is the typical case, not an edge case**
  - fraction of stored eigenstates in a degenerate eigenspace: 0.43 (N=8) → 0.74 (N=14)
  - multiplicities up to 35 within a single sector
- **Not a defect in the HSF code — a property of the spectrum**
  - options: restrict statistics to simple eigenvalues, or use a basis-independent quantity (eigenspace-averaged purity)
  - worth checking which of the other plotted rules share this

Result 5 — why the same rule gives area AND volume law

- **The three Fig. 3a states are in three very differently sized sectors**
  - single excitation |0…010…0⟩: w = 2 → |S_w| = C(N+1,2), POLYNOMIAL
  - Néel |0101…⟩: w = N → |S_w| = N+1, LINEAR
  - pair |001100…⟩: w ≈ N/2 → |S_w| exponential
- **So the premise fails: C(N+1,w) is exponentially large only near w = (N+1)/2**
  - the Néel state is a ONE-HOLE state in wall language — a wall on every bond but one, hard-core ⇒ JAMMED
- **The exact ceiling is the bipartite Schmidt rank, not ln|S_w|**
  - from w = w_A + b_cut + w_B; the LOWER limit matters at high filling
  - verified = the compressed Schmidt dimension for all w, N = 6…14
  - at N=26: single ln 92 = 4.5, Néel ln 14 = 2.6, pair 9.011 = ⌊N/2⌋ln2 exactly

Area law, area law, volume law — predicted by w alone



Left: S(t) at N=14 with the kinematic ceilings. Middle: the plateau vs N — single excitation and Néel are FLAT to N=38, the pair state grows. Right: the entanglement dome vs wall filling.

Is the sector enough? — the three answers

- **For the growth TYPE and its N-scaling: YES**
  - via the exact ceiling, not via |S_w| — knowing w already separates ln N from N
- **For PRODUCT initial states, the sector nearly fixes the value too: ±3 %**
  - six random basis states of the N=14, w=6 shell: plateaus 2.26 … 2.39
  - Fig. 3a uses product states ⇒ its three curves are set by three integers w
- **Over ALL states of a shell: NO**
  - the stationary states of the same shell — C(⌈N/2⌉, w/2) = 35 at N=14, w=6 —
  - have S(t) CONSTANT to 1e-15, at a value ABOVE the product plateau (3.5 vs 2.4)
  - a 35-dimensional family with zero entanglement growth inside a volume-law sector
- **The dome's shape is the ceiling's shape; the dynamics supplies only an O(1) fill factor ≈ ½**
  - the same sub-thermal fraction as the HSF entropies — as the degenerate spectrum predicts

Open items

- **Identify the commutant**
  - the 3^m law forces an exponentially large commutant ($\Sigma\, m_\lambda^2$)
  - its generators would prove the law AND make the block-resolved spacing statistics exact rather than a lower bound
- **Derive the two closed forms**
  - dim $\mathrm{Fix}(U_w) = C(\lceil N/2 \rceil, w/2)$, and a w-resolved version of the 3^m law
- **The partner-shell intertwiner**
  - for odd N the shells w and $N+1-w$ have identical spectra (1.2e-14),
  - but the bond particle-hole map does NOT commute with U — unidentified
- **C150 on the ring**
  - kinematics solved; the quantum layer is obc0 only. The 2-to-1 wall map gives the bond frame a Z_2 gauge redundancy
- **Frontier**
  - N = 33 needs a 64-bit label array (68.7 GB), or a two-pass compact relabelling

Reproduce

- **Closed forms, wall bijection, charge conservation, engine cross-check**
  - python -m qca_fragmentation.c150 --verify
- **Frontiers**
  - python -m qca_fragmentation.run_rule --rule 150 --N 21-22 --bc obc0
  - python -m qca_fragmentation.c150 --frontier 21-32 --bc obc0
- **Quantum layer, level statistics, entanglement, HSF check**
  - python -m qca_fragmentation.scaling.c150_report --quantum
  - python -m qca_fragmentation.quantum.rule150_levels --run
  - python -m qca_fragmentation.quantum.rule150_entanglement --run
  - python -m qca_fragmentation.quantum.hsf_compare --all
- **Report and data**
  - reports/pdf/R8_c150.pdf (18 pp) · analytics/c150_*.json
  - full test suite: 403 tests

A Tier-2 view — C150 as an open system

- Tier 2 studies the full 256-rule family as CHANNELS: each neighbour pattern carries I, V (Hadamard), D (reset to 0) or E (reset to 1)
  - C150 = (I, V, V, I) is one of the 16 unitary members — the D/E-free corner
- **Three questions the sector graph cannot answer, and the channel formalism can**
  - how large is the algebra of conserved operators? (slide 20's first open item)
  - what survives when the rule is made dissipative — are the sectors fragile?
  - is the protected coherence a decoherence-free subspace, i.e. usable as a code?
- **Everything below is an INDEPENDENT re-derivation**
  - separate repository, separate engine (sector_analysis_tier2); obc0 throughout
  - marked results are exact over $\mathbb{Q}(\sqrt{2})$ — rational Gaussian elimination, no eigensolver, no tolerance
  - first cross-check: the Tier-2 SCC code reproduces #sectors = $\lfloor (N+1)/2 \rfloor + 1$ for N = 3…12, and the pbc counts, with no shared code

Dictionary — sectors ↔ open-system language

Tier 1 (unitary)	Tier 2 (channel Φ)	C150, obc0
Krylov sector	terminal SCC of the support graph	every SCC is terminal
— (no counterpart)	transient SCCs → condensation DAG	EMPTY: the DAG has no edges
#sectors	n_recurrent	$\lfloor (N+1)/2 \rfloor + 1$
Fix(U)	$\lambda = 1$ eigenspace of Φ	dim 2^m, $m = \lceil N/2 \rceil$
commutant of U	fixed-point algebra; dim = $\Sigma\, m_\lambda^2$ = Cesàro rank	6^m (new — see next slide)
spectral degeneracy	protected coherence: gap = $\Sigma\, m_\lambda^2$ − n_recurrent	33 at $N = 4$; $6^m - \lfloor (N+1)/2 \rfloor - 1$
— (no counterpart)	transient dark states (R-T11)	none — there is no transient region

C150 is the degenerate case of the Tier-2 machinery: a channel with no transient region at all. That is exactly why every attractor question for it collapses to a question about the spectrum of U.

Open item closed (1) — the commutant

- **The commutant of U is the fixed-point algebra of the channel: dim = $\Sigma_\lambda m_\lambda^2$ = the Cesàro rank**
  - measured from the exact multiplicities, N = 3…11, and matched to a formula
- **dim C(U) = 6^m for even N, $(6^m + 2^m)/2$ for odd N, $m = \lceil N/2 \rceil$**
  - N = 3…11: 20, 36, 112, 216, 656, 1296, 3904, 7776, 23360 — exact, no fit
  - independently confirmed over $\mathbb{Q}(\sqrt{2})$ at N = 4: dim ker(L − 1) = 36 = 6^2 — tolerance-free, no eigensolver
- **Growth $(\sqrt{6})^N = 2.449^N$ — far above the 2^N of the Hilbert space**
  - the naive Cauchy–Schwarz floor from the 3^m law alone is only $4^N/3^m = 2.309^N$, so the degeneracy is even more structured than the count implies
  - only $\lfloor (N+1)/2 \rfloor + 1$ of those dimensions are diagonal (the wall charge) — the commutant is overwhelmingly OFF-diagonal
- **Protected coherence: gap = $6^m - \lfloor (N+1)/2 \rfloor - 1$**
  - 33 at N = 4, 212 at N = 6, 1291 at N = 8, 46 649 at N = 12

Open item closed (2) — the full spectrum

- **Spectral factorisation (obc0). $m = \lceil N/2 \rceil$; $M(N) = m \times m$ tridiagonal, off-diagonal ¼, diagonal ½, with**
  - $M_{00} = 1/\sqrt{2} - 1/4$ (all N) and $M_{(m-1,m-1)} = 3/4$ (odd N only)
  - $\cos \varphi_i$ = eigenvalues of $M(N)$
  - $\mathrm{spec}(U) = \{ \Sigma_i\, \varepsilon_i \varphi_i \}$, $\varepsilon \in \{-1, 0, +1\}^m$, multiplicity $2^{\#\{\varepsilon_i = 0\}}$
  - odd N: only sign patterns with an EVEN number of non-zero ε occur
- Equivalently, for even N a quantisation condition: $\sin((m+1)k) = (2\sqrt{2} - 3)\cdot\sin(mk)$, $\cos \varphi = \cos^2(k/2)$
  - the boundary defect $2\sqrt{2} - 3 = -(\sqrt{2} - 1)^2$ is the obc0 edge; the ¾ is a second defect at the far end, present only for odd N
- **Verified N = 2…12 against the engine**
  - all 729 distinct eigenphases at N = 12 reproduced from SIX numbers: max phase error 5×10^{-9}, multiplicities exact
  - it DERIVES the three laws that were stated separately: #distinct = 3^m (the all-patterns count), dim Fix(U) = 2^m (the all-zero pattern), and dim Fix(U_w) = C(m, w/2) (that pattern's split over wall shells)

Is C150 free after all? The two tests differ

- **The tension, stated plainly**
  - Result 4a: C150 is NOT free. This law: the spectrum factorises over ⌈N/2⌉ modes, which is what a free model does.
- **Resolution — they test different freeness**
  - Result 4a asks whether the w-wall block is the w-th antisymmetric power of the ONE-wall block: free, NUMBER-CONSERVING walls. That is refuted, and stands.
  - this law's modes are not wall occupations: each carries $\varepsilon \in \{-1, 0, +1\}$ (a ± pair), and there are ⌈N/2⌉ of them, not N+1
  - direct evidence: the $w = 2$ shell at $N = 6$ (21 states) contains 0-mode, 1-mode AND 2-mode phases — 3 + 6 + 12 — so mode number ≠ wall number
- **Honest summary**
  - C150 is not a free wall gas, but its Floquet spectrum is exactly that of ⌈N/2⌉ particle–hole-paired modes — spectral freeness without wall-number freeness
  - that is what makes the degeneracy structural rather than accidental, and it is the mechanism behind Result 4c's RMT exclusion
  - open: exhibit the mode operators. They generate the commutant, and a 4-level block $\{\varphi, 0, 0, -\varphi\}$ per mode gives 6^m immediately

C150 under dissipation — the neighbours

rule	tuple	recurrent classes	Cesàro rank	gap	transient dark	reading
150	(I, V, V, I)	3	36	33	—	C150 itself
146	(D, V, V, I)	1	1	0	0	total collapse, no coherence
151	(E, V, V, I)	1	5	4	0	one 5-state class: period-3 QLC
148	(I, D, V, I)	3	9	6	0	3-dim DFS = 3 sectors
134	(I, V, D, I)	3	9	6	0	3-dim DFS = 3 sectors
158	(I, E, V, I)	3	15	12	8	DFS + dark states
214	(I, V, E, I)	3	15	12	8	DFS + dark states
22	(I, V, V, D)	2	13	11	7	R-T11 rule: vacuum + same QLC
182	(I, V, V, E)	2	17	15	4	vacuum + a 10-state QLC

N = 4, obc0; all entries exact over $\mathbb{Q}(\sqrt{2})$. Note the W22 row: the rule whose transient dark states forced the Tier-2 gap definition to be corrected — is C150 with a single symbol changed. The two middle-slot D-mutations preserve the sector count exactly; the E-mutations add transient dark states on top.

The sectors survive dissipation as a DFS

- **W148 = (I, D, V, I) and W134 = (I, V, D, I): each C150 sector collapses to ONE pointer state, and every coherence between them survives**
  - exact block dimension [n] with $n = \lfloor (N+1)/2 \rfloor + 1$ = the C150 sector count, verified N = 3…7 for both rules; gap = $n(n-1)$
  - N = 3,4 → 3-dim (gap 6); N = 5,6 → 4-dim (gap 12); N = 7 → 5-dim (gap 20)
- **Explicit, at N = 5: DFS = span{ |00000⟩, |10000⟩, |10100⟩, |10101⟩ }**
  - one maximally packed representative per wall number w = 0, 2, 4, 6 — left-packed for W148, right-packed for W134, the two being boundary mirror images. The dissipation picks a canonical member of each C150 sector and keeps the coherence between them
- **Reading: the wall charge becomes an error-protecting charge**
  - a logical qudit of dimension $\lfloor (N+1)/2 \rfloor + 1$, growing LINEARLY in N, protected passively — no gates, no syndrome extraction
  - same mechanism as W232 in Tier-2's QEC chapter (R-T12; cf. Guedes, Winter & Müller, arXiv:2309.03608): a classical CA charge promoted to a dissipative code space
  - caveat: this is protection against the automaton's OWN dissipation. No threshold against extrinsic noise has been computed — that is the next test.

The ring, and the updated open list

- **C150 on pbc — kinematics in closed form (Tier-2 SCC engine, $N = 3\ldots12$)**
  - $|S_w| = 2\cdot C(N, w)$ for even w with $0 < w < N$; the factor 2 IS the Z_2 gauge redundancy — the two spin states of a bond string are the global flip
  - $w = 0$ gives two frozen singletons (all-0, all-1); for even N, $w = N$ gives two more (the two Néel states)
  - #sectors = $N/2 + 3$ (even N), $(N+3)/2$ (odd N) — checked $N = 3\ldots12$
- **Closed by this chapter**
  - the commutant dimension (6^m), and its $(\sqrt{6})^N$ growth
  - the 3^m law, $\dim \mathrm{Fix}(U) = 2^m$ and $\dim \mathrm{Fix}(U_w) = C(m, w/2)$ — all three now follow from one factorisation
- **Still open**
  - the explicit mode operators; the partner-shell intertwiner; whether the odd-N second defect ¾ has a boundary interpretation
  - does the W148/W134 DFS dimension keep tracking $\lfloor (N+1)/2 \rfloor+1$ beyond $N = 7$, and does it hold on the ring?
  - extrinsic-noise threshold for the inherited code — connects this deck to the QEC literature